

# JUNXIAN WU

## Senior AI Engineer | Computer Vision & Edge AI

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### PROFILE

Senior AI Engineer delivering computer vision and computational photography for mobile camera systems. Experience spans model development, quantization, C++/Android integration, visual-quality evaluation and cross-team release delivery. Independent projects explore agent orchestration and local image-analysis workflows.

### PROFESSIONAL EXPERIENCE

#### Senior AI Engineer | Black Sesame Technologies (Singapore)

Jan 2025 - Present

- Own camera AI delivery from product requirements and field feedback through prototypes, hard-case evaluation and release handoff with product, camera tuning, runtime and QA teams.
- Lead portrait-imaging development across matting, monocular depth and GPU rendering, investigating subject boundaries, visual artifacts and device behavior.
- Develop real-time composition workflows combining object detection, optical-flow tracking, camera guidance and application state.
- Drive PyTorch-to-device integration through quantization, TFLite/ONNX export validation, native C++/Android pipelines and on-device debugging.

#### AI Engineer | Black Sesame Technologies (Singapore)

Aug 2022 - Jan 2025

- Productized portrait segmentation, matting and monocular depth models for mobile camera pipelines.
- Built training-to-deployment workflows with quantization, export checks and regression comparisons to validate model replacements.
- Integrated models into C++/Android runtimes and developed depth-aware bokeh with visual debugging, golden cases and batch release-review tooling.

### RESEARCH EXPERIENCE

#### Research Intern | National University of Singapore | Dec 2021 - Jul 2022

Computer-vision research for construction-site safety; co-authored a peer-reviewed Automation in Construction paper.

#### Research Assistant | Shandong University | Sep 2019 - Jun 2021

Applied computer vision and machine learning to interdisciplinary psychology research, datasets, experiments, and reporting.

### SELECTED PUBLIC SYSTEMS

**Reliable Agent Workflow:** Scripted-provider Python reference for retries, fallback, output validation and human-review gates, with trace-based regression tests. [Code and tests](#)

**Photography Mentor:** Browser-local image diagnostics, source-linked retrieval and practice tracking using deterministic analysis and coaching rules. [Live application](#)

### TECHNICAL STACK

- **AI and vision:** PyTorch, TensorFlow, OpenCV, segmentation, matting, monocular depth, detection, optical flow, local retrieval
- **Runtime and quality:** Python, C++, Android/NDK, TFLite, ONNX, TensorRT, OpenGL/GLSL, Docker, quantization, hard cases, traces, release gates

### EDUCATION

Master of Technology, Intelligent Systems | National University of Singapore, ISS | 2021 - 2022

Bachelor of Engineering, Software Engineering | Shandong University | 2017 - 2021 | Outstanding Graduate

### PUBLICATION

**Co-author**, "Automated classification of 'cluttered' construction housekeeping images through supervised and self-supervised feature representation learning," *Automation in Construction*, 2023. [doi.org/10.1016/j.autcon.2023.105095](https://doi.org/10.1016/j.autcon.2023.105095)